

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Artificial Intelligence

2. **Subject Name:** The Essentials of Machine Learning **L-T-P: 3-0-0** **Credits: 3**

3. **Pre-requisites:** None

4. **Syllabus and reference books**

Syllabus:

- **Supervised Learning:** K-Nearest Neighbour, Decision Tree, Linear, Regularized and Logistic Regression, Support Vector Machine, Perceptron, Ensemble Classification.
- **Unsupervised Learning:** Hierarchical and flat clustering. Bayesian classifiers, Gaussian Mixture Models. Dimensionality Reduction
- **Deep Learning models:** Feed-forward Neural Nets, CNNs, Language Models: RNNs and Transformer

Reference Books:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 4th Edition, 2020
2. Pattern Recognition for Machine Learning, Christopher M. Bishop, Springer, 2006
3. Neural Networks and Deep Learning: A Textbook, Charu C Aggarwal, Springer, 2018
4. Deep Learning, Aaron Courville, Ian Goodfellow, Yoshua Bengio, MIT Press, 2016.

5. **Lecture-wise break-up:**

No	Module	Topic	Lecture-Hours
1	Introduction	Types of learning, Data Representation, Case studies	2 hours
2	Regression	Linear and regularized regression	2 hours
3	Simple Classifiers	K-Nearest Neighbour, Decision Tree, Logistic Regression	4 hours
4	Basic Optimization	Gradient Descent and its variants, Constrained Optimization	4 hours
5	Max-margin Classifier	Linear Separability, Support Vector Machine, Kernel Methods in SVM	3 hours
6	Ensemble Classification	Bagging, Random Forest, Boosting, Adaboost	3 hours
7	Unsupervised Learning	Flat, hierarchical and agglomerative clustering, spectral clustering	4 hours
8	Probabilistic Approaches	Naïve Bayes Classifier, Gaussian Mixture Models	3 hours
9	Dimensionality Reduction	Singular Value Decomposition (SVD), Principal Component Analysis (PCA)	3 hours
10	Perceptron and Feedforward net	Perceptron, Feed-Forward Neural Network, Backpropagation Algorithm	4 hours
11	Vision Models	Building blocks of convolutional neural nets, Basic CNN, VGG16, Residual networks	3 hours
12	Language Models	Recurrent neural nets (RNN, LSTM), Introduction to Transformer	4 hours